Sci. & Soc. 23(2) 179-188, 2025 ISSN 0973-0206

Predictive Machine Learning for Natural Disasters

Elizabeth Poulose and Preethy George

Department of Computer Science, Nirmala College Muvattupuzha (Autonomous), Kerala

Received on 25 September 2025; Accepted on 22 October 2025

Abstract

In the midst of continuing climate changes and increasing upturns in natural disasters change, it is imperative to achieve strong resilience against these natural calamities. This chapter delves into the dynamic crosswalk of machine learning and natural disasters, revealing how these technologies can revamp disaster management. In spite of accentuating challenges in natural disasters like cyclones, flooding, forest fires, tremors, machine learning develops into an effective solution and equipping proactive strategies in addition to conventional reactive methods.The story unfolds by tracing the advancement of disaster management, emphasis the transformative influence of machine learning on early warning systems. Going beyond just forecasting, the conversation also covers the crucial part that machine learning plays in enhancing the efficiency of response and recovery actions effectively distributing sources and promoting the planning for recovery. A machine learning method incorporates Naive Bayes (NB) methods, Decision Trees (DT), Support Vector Machine (SVM), Random Forest (RF), Regressions and K-nearest neighbour (KNN), clustering algorithm. The comprehensive objective of the analysis presented in this paper is to manifest the potential handling of data about natural disasters and their significance for societal safety and industrial management.

Keywords: Proactivestrategies, Response and recovery, Reactive methods